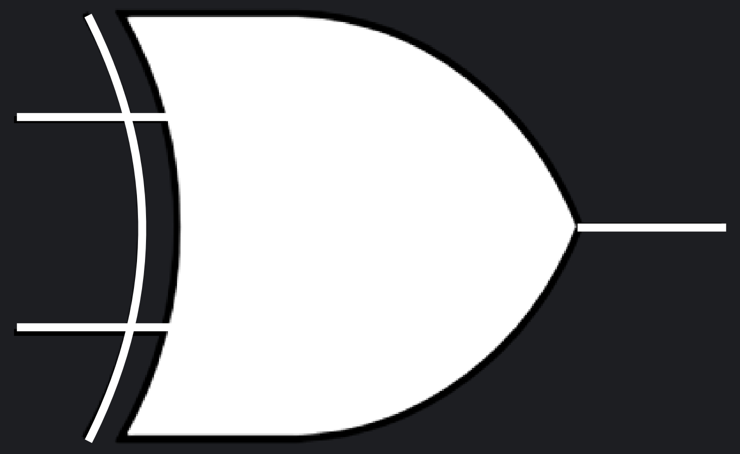
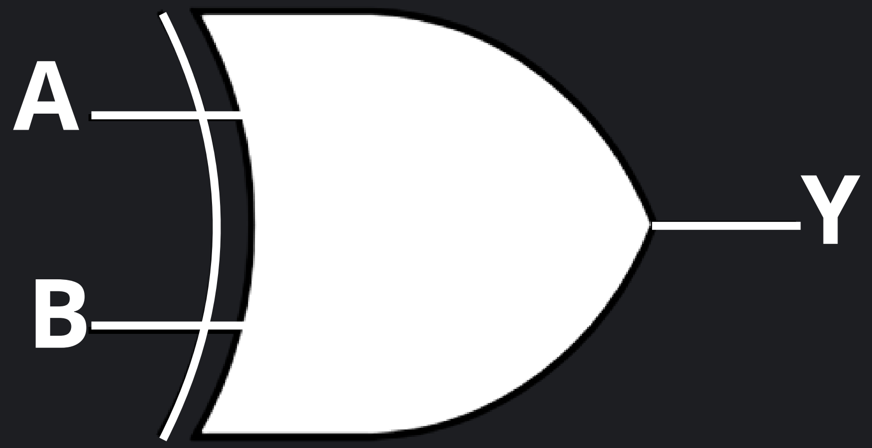
XOR GATE



What is a XOR Gate?

An XOR gate is a basic logic gate that produces a HIGH (1) output when its inputs are different. The output is LOW (0) when all inputs are the same (either all LOW or all HIGH).

Truth Table



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Function of the XOR Gate

An XOR gate performs a logical Exclusive OR (XOR) operation on its inputs.The main function of an XOR gate is to combine two or more input signals and produce an output signal according to this rule:

Output = 1 when the inputs are different

Output = 0 when the inputs are the same (either all 0 or all 1)

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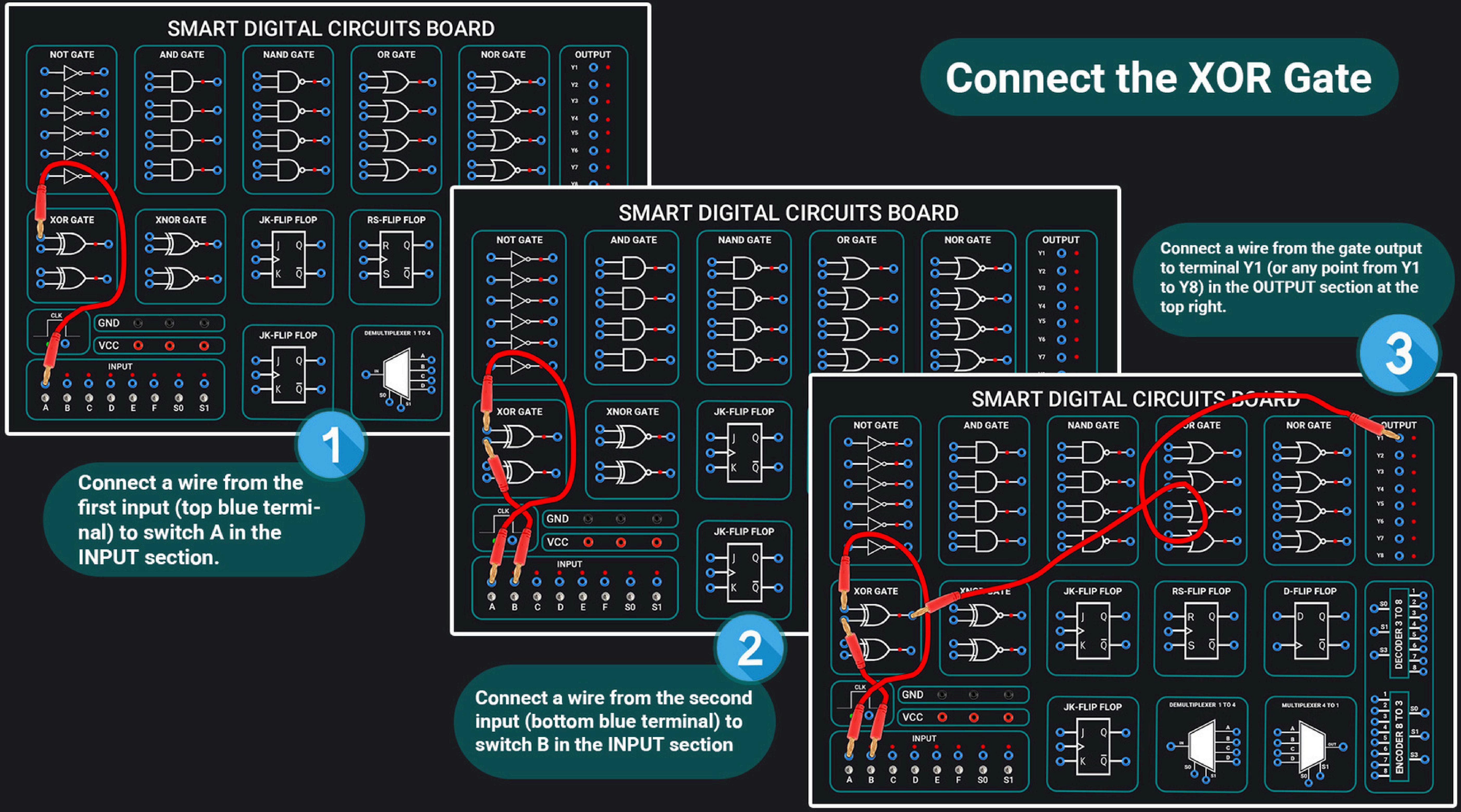
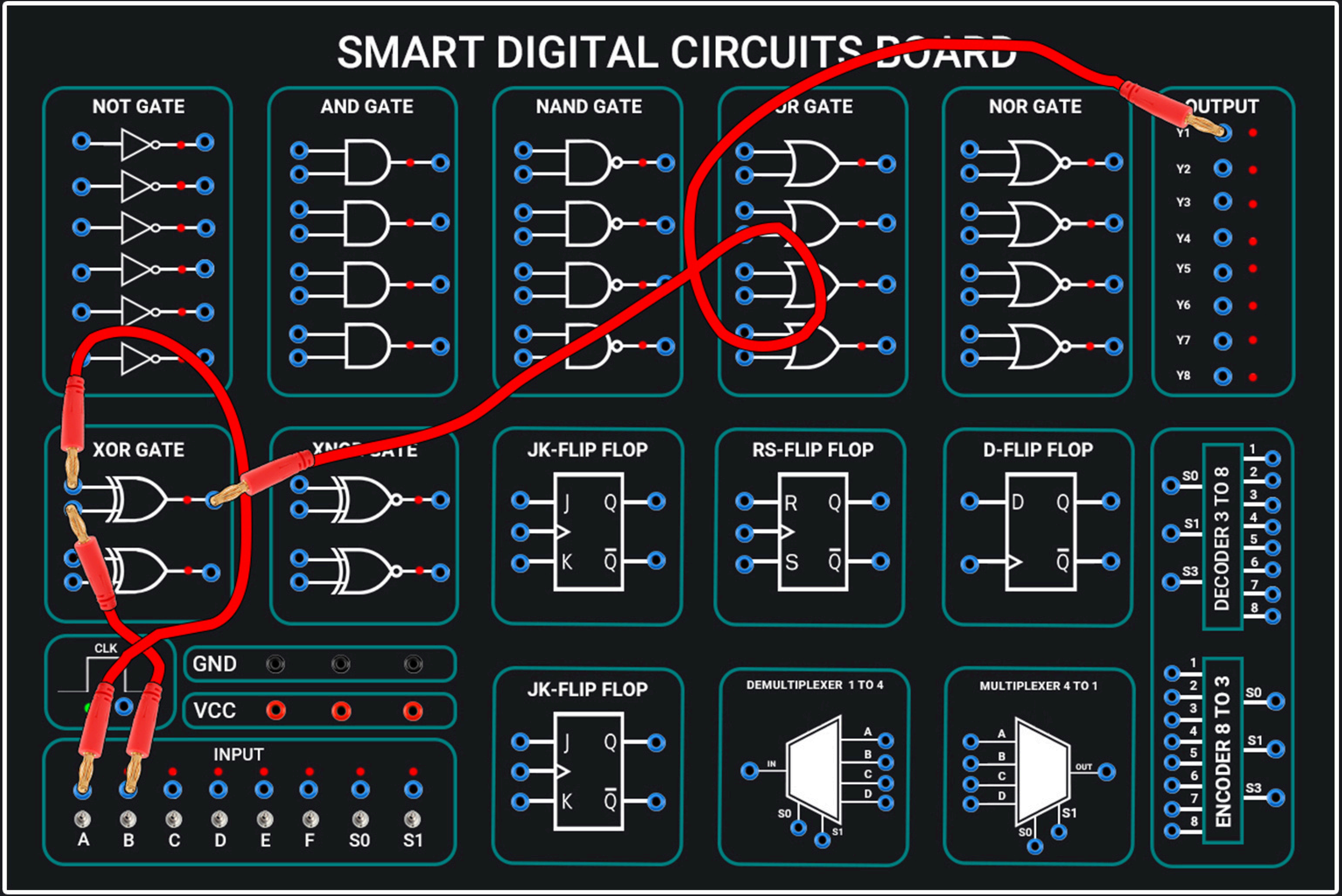
Connect the XOR Gate



1. Connect a wire from the first input (top blue terminal) to switch A in the INPUT section.

2. Connect a wire from the second input (bottom blue terminal) to switch B in the INPUT section

3. Connect a wire from the gate output to terminal Y1 (or any point from Y1 to Y8) in the OUTPUT section at the top right.


SMART DIGITAL CIRCUITS BOARD
NOT GATE
AND GATE
NAND GATE
NOR GATE
OUTPUT
Y1
Y2
Y3
Y4
Y5
Y6
Y7
Y8
XOR GATE
JK-FLIP FLOP
RS-FLIP FLOP
D-FLIP FLOP
DECODER 3 TO 8
S0
S1
S3
1
2
3
4
5
6
7
8
J
Q
K
R
S
D
CLK
GND
VCC
JK-FLIP FLOP
DEMULTIPLEXER 1 TO 4
IN
A
B
C
D
S0
S1
MULTIPLEXER 4 TO 1
OUT
ENCODER 8 TO 3
INPUT
A
B
C
D
E
F
S0
S1